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Data Science Project

# Run pace predictor

## Final project

15 Jun 2026

## Abstract

This project presents the development of a personalized run pace predictor built from an individual Garmin Connect training history. The system addresses a key limitation of commercial sports watches, which provide race time estimates only for standard distances and do not account for elevation gain.

The pipeline covers four phases: data acquisition via the Garmin Connect API, data cleaning and feature engineering, machine learning modelling, and a prediction interface. Three models are trained and compared — Ridge Regression, Random Forest, and XGBoost, using a dynamic 80/20 temporal split to ensure long-term reliability. Feature engineering plays a central role, combining TRIMP-based training load indicators (ATL, CTL, TSB) with Riegel-inspired fatigue features that capture the non-linear relationship between distance, elevation, and pace degradation.

XGBoost achieves the best performance, with a mean absolute error of 3 minutes and 20 seconds and an  $R^2$  of 0.973 on the test set. For races exceeding the training distribution, the model falls back to Riegel's extrapolation formula. The results are exposed through a Streamlit web application that allows the athlete to input race parameters and instantly receive a predicted finish time, average pace, and physiological context for any planned race date.

The project shows that a highly accurate and personalized prediction tool can be built from a single athlete's data, and identifies clear directions for future improvement, including richer long-distance data and additional physiological signals.

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## 1. Introduction

### 1 Project Objectives

Current sports watches record a wide range of data during activities. They allow athletes to track their progress. Some watches also use this data to offer personalized training sessions aimed at improving athletes' abilities. Based on all this data, some watches are even capable of predicting potential run times in advance.



*Figure 1: Garmin race predictor*

Unfortunately, the estimated times are only for "standard" distances and do not take into account elevation changes or other factors that could affect race times.

**The objective of this work will therefore be to create a model, that based on training data as well as the parameters of an upcoming run, can estimate a potential run time.**

## 2. Project pipeline

The pipeline of this project is split between 3 big phases.

In the first phase, **Data acquisition**, the raw data is downloaded from the Garmin API, then filtered and cleaned. Next, features useful for modeling are created, and the dataset is saved. Each time the script runs, the latest data is downloaded from the API. Once the data is available, exploratory data analysis is performed.

The **modelling phase** aims to predict `duration_s` (total race time in seconds) from the 18 engineered features. A dynamic 80/20 temporal split is applied — the oldest 80% of runs form the training set, the most recent 20% the test set — ensuring the pipeline remains valid over time without data leakage. Three models are trained and compared: Ridge Regression as a linear baseline, Random Forest, and XGBoost. All three are evaluated on MAE, RMSE, and  $R^2$ .

The **prediction phase** exposes the trained XGBoost model through a `predict_course()` function that takes four inputs: distance (km), elevation gain (m), race type (road or trail), and target date. At inference time, the function automatically recomputes the athlete's current fitness state (ATL, CTL, TSB) at the specified date and retrieves the latest recorded VO2max, ensuring the

prediction reflects the physiological context at the time of the race. If the requested distance or elevation exceeds the maximum values observed in the training set (68.5 km and 1,864 m respectively), the model flags an extrapolation warning and falls back to Riegel's formula, which handles long-distance fatigue more reliably outside the training distribution.

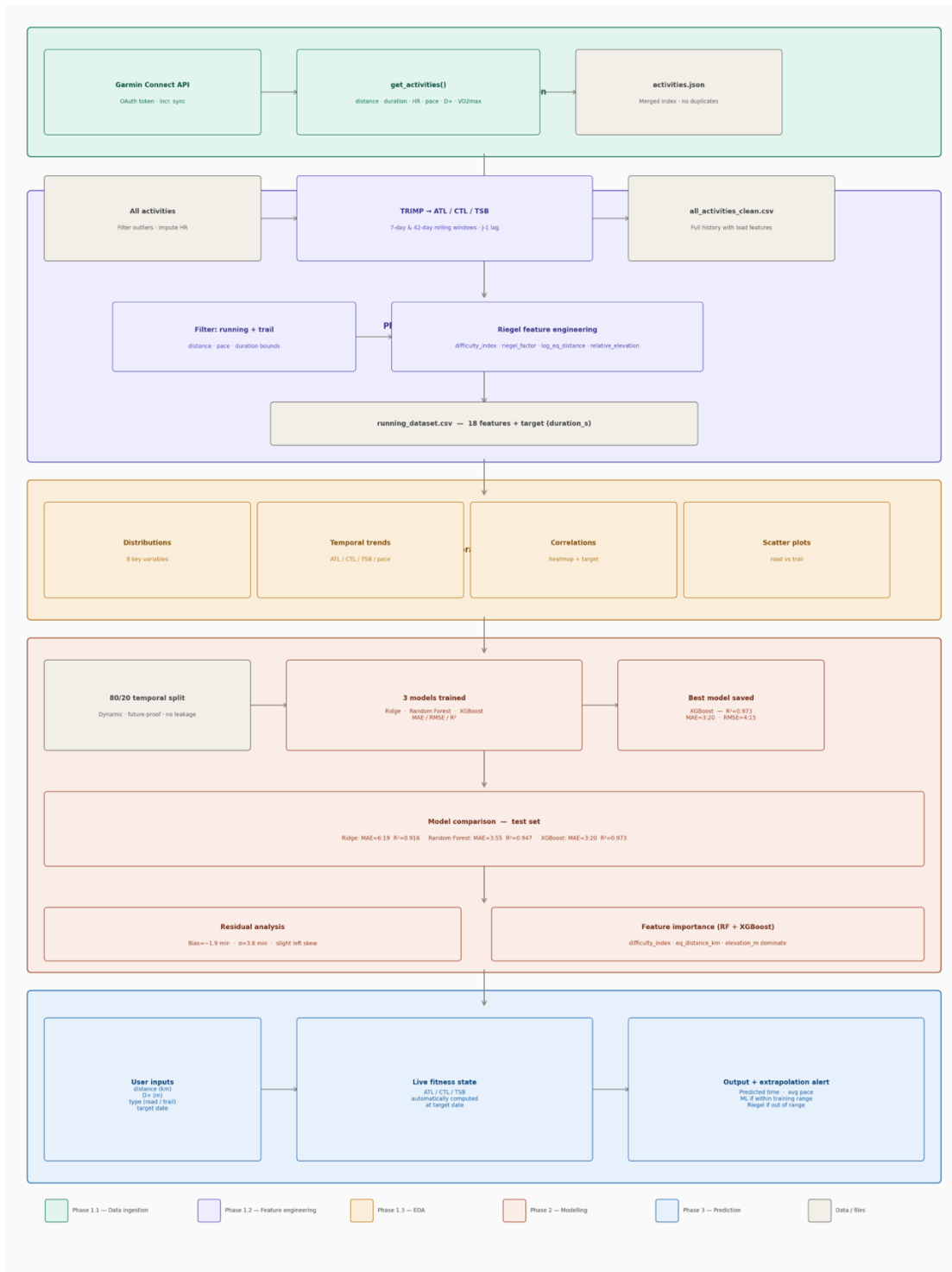


Figure 2: Project pipeline

### 3. Data

#### 3.1 Source of the data

For this project, the data analyzed will be the data generated and collected by the author's watch during his workouts.

Each workout recorded on the watch is stored in the author's Garmin Connect account. The primary objective of this project is therefore to retrieve the data, load it into a database, and then update that database over time so that all recorded activities are always available.

#### The Importance of Up-to-Date Data

In sports, "historical" data reflects past performance levels and becomes less informative as training progresses. Recent training data reflects the athlete's current physiological condition. It is therefore important to have historical data available to track performance trends and recent data to assess the athlete's current condition in preparation for an upcoming event.<sup>1</sup>

#### 3.2 Accessing and loading the data

The connection to the Garmin Connect account is made via the "garminconnect" library. From there, activities and their metadata are downloaded. An incremental synchronization system ensures that only activities that have not been previously uploaded are downloaded. At this stage, all recorded activities are downloaded, with the exception of dive data.

The metadata for the activities is as follows:

- Distance
- Duration
- Average heart rate (HR)
- HR max
- Pace
- Elevation gain
- VO2max
- Aerobic/anaerobic effects
- Cadence
- Calories
- Training load

GPS data for activities is intentionally omitted due to the large amount of data it would generate.

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<sup>1</sup> (Frank Imbach, 2022)

The activities are then saved in a ".json" file.

### 3.3 Cleaning the data

Once the data has been collected, they need to be cleaned :

- Abnormal activities (duration <1 minute, distance <100 m) are excluded.
- If the heart rate is missing, it is replaced by the median value for that type of activity.
- For run data, a specific filter is applied to keep the runs
  - a distance of between 1 and 100 kilometers;
  - lasting between 5 and 720 minutes;
  - a pace between 200 and 1,200 seconds per kilometer.

All activities are supervised, but not all of them have the same purpose:

- All activities are used to calculate the training load.
- Running and trail running activities form the dataset used for modeling.

### 3.4 Features engineering

Once the data is ready, new features are created to determine the type of training and to model the gradual slowdown over distance during a run.

#### Training load

The workload for each session is quantified using TrainingIMPulse (TRIMP), a method developed by Eric Banister in 1991 that combines exercise duration with the intensity of the effort, as estimated from heart rate<sup>2</sup>.

Based on TRIMP, three additional load indicators are calculated using moving windows and linked to each run (the calculated values are the J-1 values to prevent data leakage).

| Indicator                     | Window  | Signification |
|-------------------------------|---------|---------------|
| Acute training load (ATL)     | 7 days  | Acute fatigue |
| Chronic Training Load (CTL)   | 42 days | Chronic form  |
| Training Stress balance (TSB) | CTL-ATL | Freshness     |

*Table 1: Training load features*

In addition, there are three new features:

- Total load over the past 7 days
- Total load over the past 28 days

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<sup>2</sup> (Impulse, 2012)

- The number of "rest" days since the last activity

### Running-specific fatigue characteristics

To enable the model to extrapolate correctly over long distances, four additional features are created to model the gradual decrease in speed. These features are based on Riegel's formula<sup>3</sup>.

| Features         | Formulas                                         | Description                       |
|------------------|--------------------------------------------------|-----------------------------------|
| Riegel_Factor    | $\text{Distance\_km}^{1.06}$                     | Gradual slowdown                  |
| Distance_eq_km   | $\text{Distance\_km} + \text{Dénivelé\_m} / 100$ | Trail correction                  |
| Difficulty_Index | $\text{Distance\_eq\_km}^{1.06}$                 | Combine elevation gain and Riegel |
| Log_distance_eq  | $\text{Log}(\text{Distance\_eq\_km})$            | Nonlinear relationship            |
| Denivele_relatif | $\text{Dénivelé\_m} / \text{distance\_km}$       | Relief intensity                  |

*Table 2: Run-specific fatigue characteristics*

### Final data set

The final dataset thus includes 18 input features<sup>4</sup> and one target variable: "duration\_s". Variables that are unknown prior to a future run, such as "pace\_s\_km" and "avg\_hr", are excluded from the predictive features.

## 4. Exploratory data analysis (EDA)

The dataset includes 446 activities carried out between December 3, 2022, and June 10, 2026.

Of these activities, 218 are road runs and 45 are trail runs.

### General running metrics:

Distance: 1.5 – 68.5 km (average 9.9)  
Duration: 7 – 448 minutes (average 54)  
Elevation: 9-1864 meters (avg 217)  
Pace: 258/839 s/km (average 321)

Among all the run, no features are missing for any event and no run had to be removed.

### 4.1 Some major distributions

#### Duration and distance

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<sup>3</sup> (Walker, 2025)

<sup>4</sup> See the appendix for a complete list of features.



These two graphs show the relationship between the number of runs and duration/distance. For both variables, there is a strong right-skewed distribution with a few outliers (long runs). These “extreme” values will serve as extrapolation cases for the model.

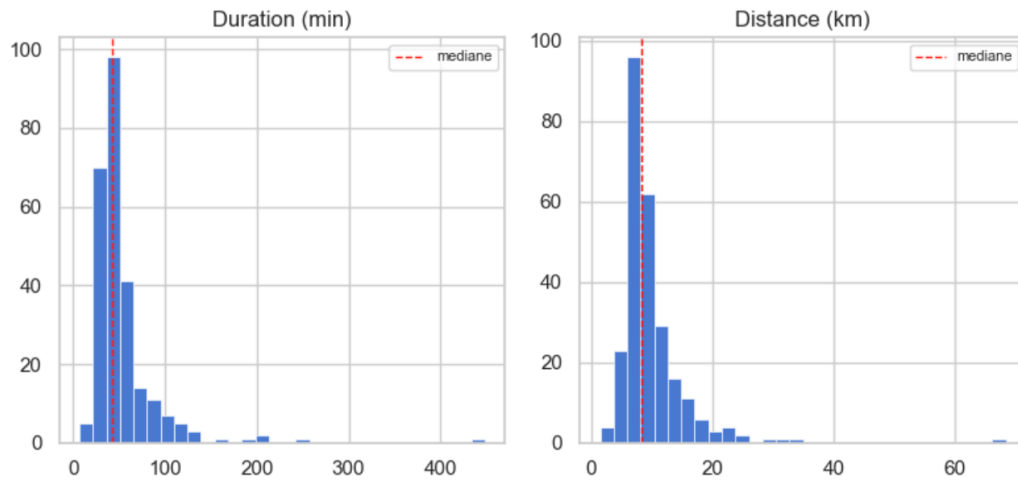


Figure 3: Duration and distance

## Elevation

The elevation gain distribution is concentrated, and the median is very low. This indicates that most runs take place on flat terrain.

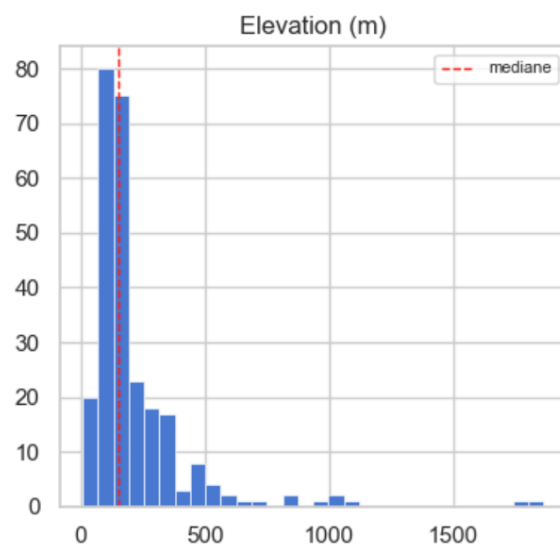


Figure 4: Elevation

## Heart rate

The heart rate follows a near-normal distribution, which seems to indicate a wide range of training intensities over time.

The running pace remains stable over time. The graph shows that the first trail runs appear starting in mid-2023 and become more frequent in 2024–2026. We also notice outliers with very slow paces here.

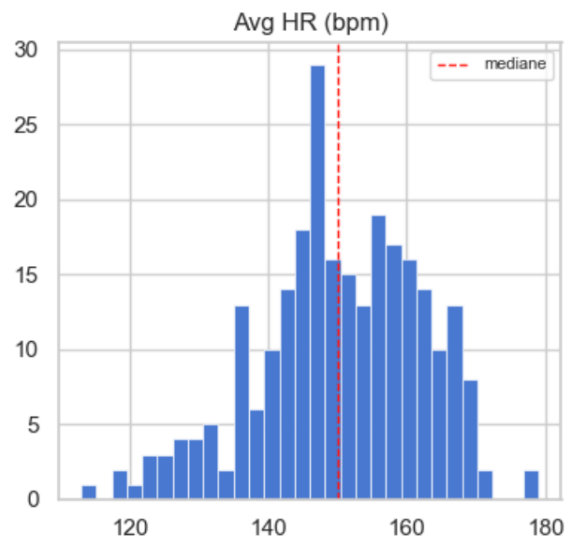


Figure 5: Average Heart rate (HR)

### Conclusion – Importance for the models

There is a strong focus on short distances, and the model will struggle to predict long distances. This justifies the addition of the “log\_distance\_eq” variable to account for a gradual slowdown during longer runs.

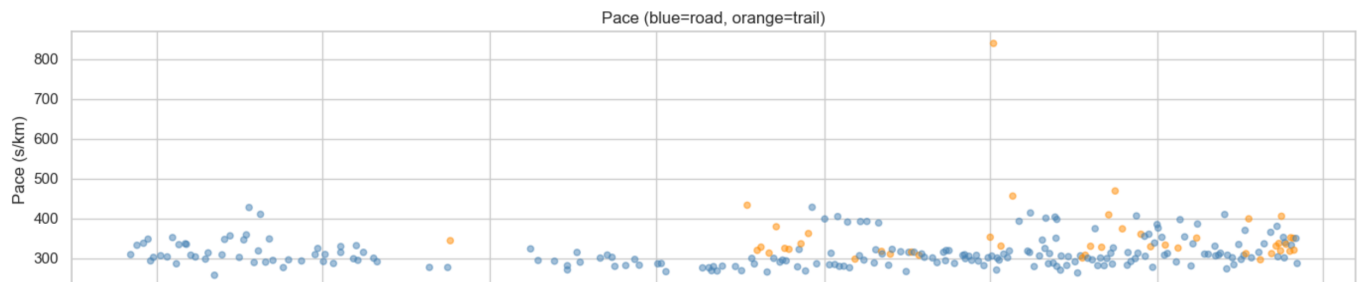


Figure 6: Pace over time

### Training load and freshness

The training load is initially moderate between 2022 and 2023, then gradually increases starting in 2024. It peaks around mid-2025 and remains at a high level through 2026.

Recovery is consistent with the training load. The “fresh” green zones are short, and runs generally take place while the body is in a state of mild accumulated fatigue.

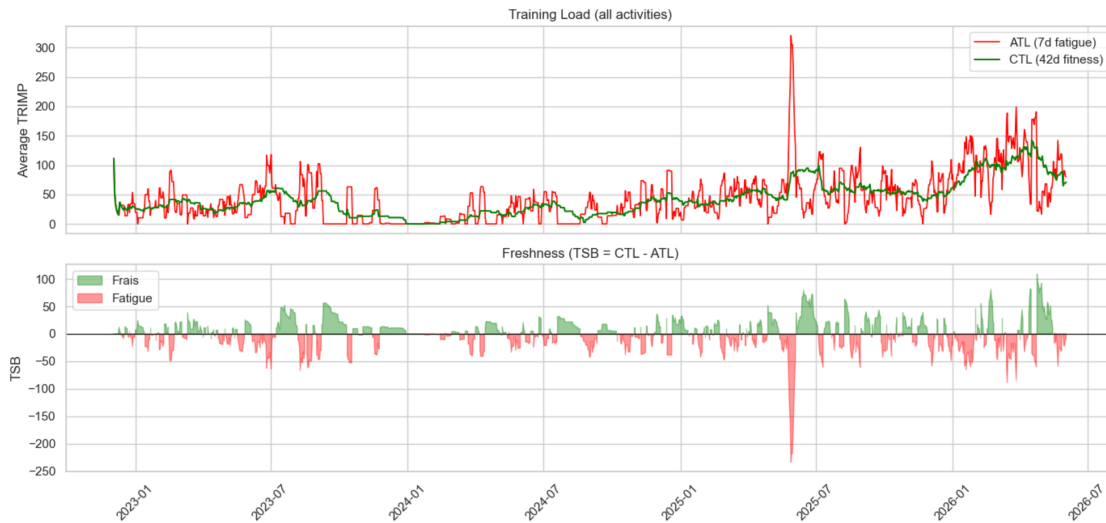


Figure 7: training load over time

These graphs appear to confirm the relevance of the training load features (ATL, CTL, TSB) in the model. The temporal variability of the training load is clear and significant. A model without these features would ignore the physiological context in which the runs are performed.

## 4.2 Correlation of variables with duration

Three groups stand out in the correlation chart:

- 1) The **eq\_distance\_km** metric, **distance in kilometers**, and **elevation** are **strongly positively correlated**, which is not surprising since duration is directly related to distance and elevation gain slows you down.
- 2) Some variables are **positively correlated, though to a lesser extent**. Some variables show a moderately positive correlation with duration that may seem counterintuitive at first glance. Pace per kilometer (**pace\_s\_km**) is a good example: one might expect faster runners to run longer, but the relationship is in fact the opposite. Longer runs are systematically performed at a slower pace which explains why a higher pace (more seconds per kilometer) correlates positively with a longer duration. This is not a physiological paradox but a mechanical consequence of the dataset structure: distance drives both duration and pace simultaneously. Similarly, the aerobic effect shows a moderate positive correlation because longer, slower efforts generate a greater aerobic stimulus.
- 3) Some variables show a slightly negative correlation. These include pace, which is higher during shorter runs, and freshness, which indicates shorter runs when one is well-rested. Finally, certain variables such as VO2max, average heart rate, or the number of days since the last activity show virtually no correlation.

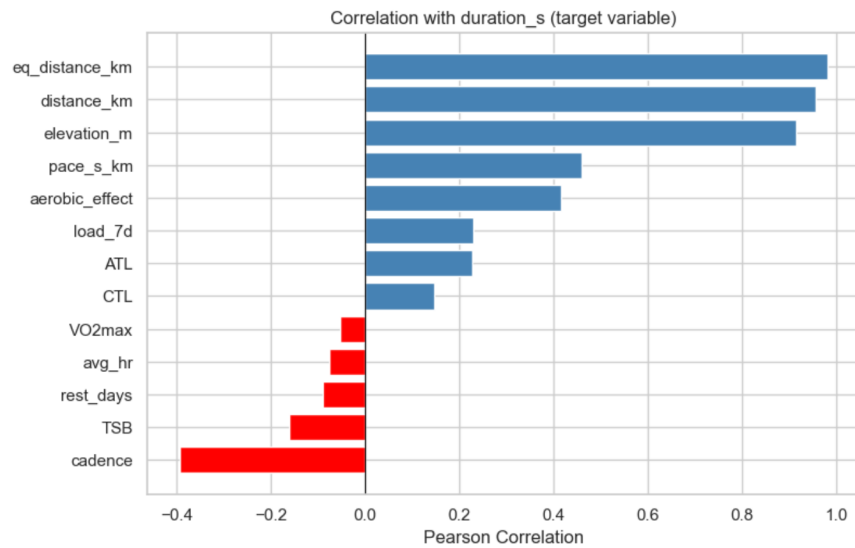


Figure 8: Correlation between features and duration

### Conclusion for the model

Distance (and its equivalent measure) accounts for approximately 95% of the variance in duration on its own. All other features play a marginal but cumulatively significant role, particularly in distinguishing between two runs of the same distance run under very different fitness conditions. This is precisely where ATL/CTL/TSB and VO2max add value, even though their individual linear correlations appear weak.

## 5. Statistical analysis

### 5.1 Models

The goal here is to create a model that best predicts a run based on the recorded data. The target variable is "duration\_s" (total time in seconds).

For validation, a dynamic time split was chosen to evolve over time. The oldest 80% of runs constitute the training set, and the most recent 20% constitute the test set. This ensures the script's long-term reliability.

#### Three models are developed and then compared

1. **Ridge regression (baseline):** This is a linear regression with L2 regularization ( $\alpha=1$ ), preceded by StandardScaler normalization. The coefficients allow for a direct interpretation of the importance of the features
2. **Random Forest:** a forest of 300 trees, with no depth constraint, and a minimum of 3 samples per leaf. Provides a measure of feature importance through permutation.

3. **XGBoost**: 400 estimators, learning rate of 0.05, maximum depth of 4, subsampling of 0.8.

### Evaluation metrics

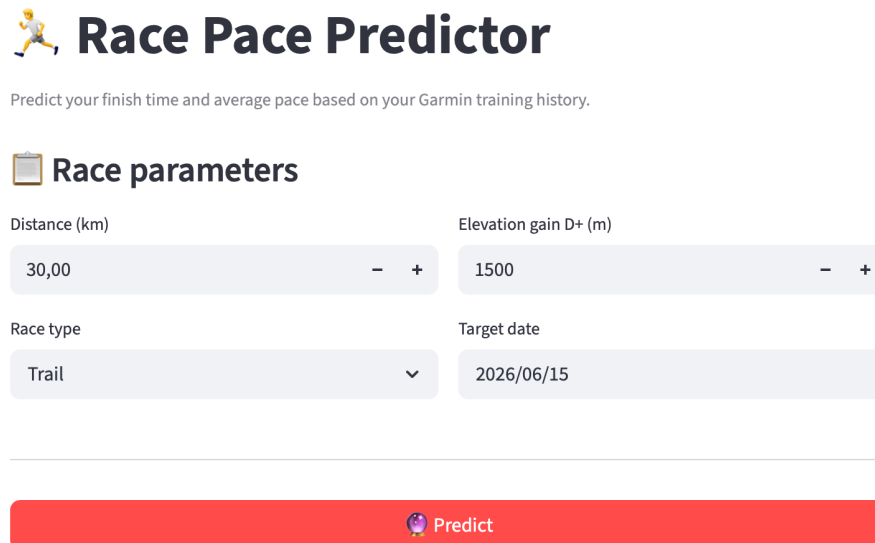
- MAE: mean absolute error in seconds (also converted to s/km).
- RMSE: root mean square error (penalizes large errors more heavily).
- $R^2$ : proportion of variance explained.

The best model is then selected and saved in order to determine a time or pace for a run based on specific parameters.

### 5.1 Output view

A lightweight web application was developed using the Streamlit framework to make the race pace predictor accessible without requiring any programming knowledge. The app is launched locally from the terminal with a single command (`streamlit run app.py`) and opens automatically in the browser.

The interface is organized in three sections. First, the user **fills in four race parameters**: the target distance in kilometers, the positive elevation gain in meters, the race type (road or trail), and the planned race date. Once the user clicks the **Predict** button, the app automatically retrieves the athlete's fitness state at the specified date by recomputing ATL, CTL, and TSB from the full activity history.



**Race Pace Predictor**

Predict your finish time and average pace based on your Garmin training history.

**Race parameters**

|               |                       |
|---------------|-----------------------|
| Distance (km) | Elevation gain D+ (m) |
| 30,00 - +     | 1500 - +              |
| Race type     | Target date           |
| Trail v       | 2026/06/15            |

 Predict

*Figure 9: upcoming run parameter*

**The results are then displayed** in two blocks. The first block shows the three key outputs predicted finish time, average pace in min/km, and the method used (ML model or Riegel extrapolation) as headline metrics. The second block displays the athlete's physiological context

on race day: ATL (acute fatigue), CTL (chronic fitness), TSB (freshness), and the latest recorded VO2max.

### Prediction

Finish time

4:21:12

Average pace

8:42 /km

Method

ML — XGBoost

### Fitness state on race day

ATL (fatigue)

109.2

CTL (fitness)

93.5

TSB (freshness)

-15.7

VO2max

57.0

↑ Fatigued ⚠️

Mild accumulated fatigue — the prediction accounts for this.

Figure 10: Prediction results

Finally, a split table **estimates intermediate times at standard checkpoints** (5 km, 10 km, half-marathon, marathon, and so on up to the target distance), giving the athlete a practical pacing reference for race day.

### Estimated split times

| Distance (km) | Split time | Pace     |
|---------------|------------|----------|
| 5.0           | 0:43:32    | 8:42 /km |
| 10.0          | 1:27:04    | 8:42 /km |
| 15.0          | 2:10:36    | 8:42 /km |
| 20.0          | 2:54:08    | 8:42 /km |
| 21.1          | 3:03:42    | 8:42 /km |
| 30.0          | 4:21:12    | 8:42 /km |

Figure 11: split times

## 6. Result discussion

### 6.1 Results

The three models produce the following results:

| Model                       | MAE   | RMSE  | R <sup>2</sup> | Pace error |
|-----------------------------|-------|-------|----------------|------------|
| Ridge regression (baseline) | 6 :19 | 7 :31 | 0.916          | +35.6 s/km |
| Random Forest               | 3 :55 | 5 :59 | 0.947          | +22.1 s/km |
| XGBoost                     | 3 :20 | 4 :15 | 0.973          | +18.8 s/km |

Table 3: Models results

Of the three models, XGBoost performs best across all metrics. With an R<sup>2</sup> of 0.973, it explains 97.3% of the variance in duration on the test set. The consistent improvement across the three models demonstrates that more complex models are better able to capture the nonlinearities of the problem.

### XGBoost residuals analysis

The predicted and actual values align well, confirming R<sup>2</sup> = 0.973. We observe that the trail running data generally follow the same trend as the road running data. The model thus generalizes correctly across both types of running despite their very different parameters. Variation increases for longer durations. Absolute errors increase with distance, even though relative errors remain stable.

With an **average bias of -111 seconds**, the model tends to slightly overestimate the actual duration. This bias is small but systematic. Since the model's performance has been steadily improving since the beginning of the training set, the data in the test set (the most recent 20% of the data) is thus above the historical average.

As for the **standard deviation of the residuals of 230 seconds**, the dispersion remains reasonable. The  $\pm 1\sigma$  confidence interval covers the vast majority of predictions within a range of  $\pm 4$  minutes.

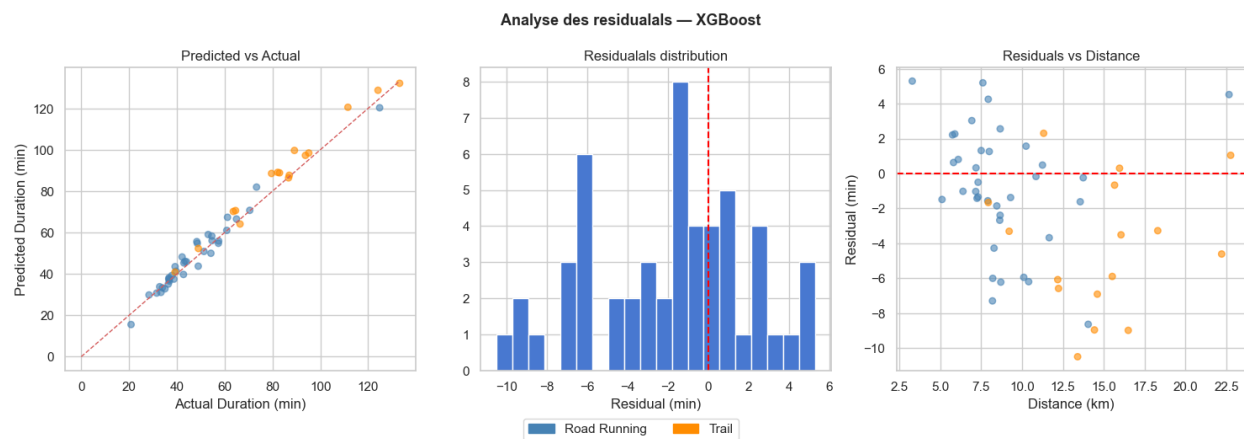


Figure 12: Residuals analysis

### Feature importance

Both models converge on the same four dominant features: `difficulty_index`, `eq_distance_km`, `elevation_m`, and `distance_km`. This consistency across two very different algorithms confirms that these features are genuinely informative rather than artefacts of a specific model. Quantitatively, `difficulty_index` ranks first in Random Forest (26%) and second in XGBoost (35%), while `eq_distance_km` ranks first in XGBoost (37%) and third in Random Forest (22%). Together, these two engineered features alone account for roughly half of the predictive power in both models.

The most notable divergence between the two models concerns `log_eq_distance`: it is the second most important feature in Random Forest (23%) but near-negligible in XGBoost (3%). This is not a contradiction — it reflects a fundamental difference in how the two algorithms handle non-linearity. Random Forest requires the log transformation to linearise the distance-duration relationship within its splits, whereas XGBoost approximates the same non-linear curve naturally through its sequential boosting mechanism, rendering the transformation redundant.

Physiological state features (ATL, CTL, TSB, VO2max, `training_load`) all fall below 1–2% individually in both models. This does not mean they are useless, but rather that the geometry of the race — distance and elevation — dominates the signal to such an extent that fitness context plays a secondary role given the current dataset size and composition.

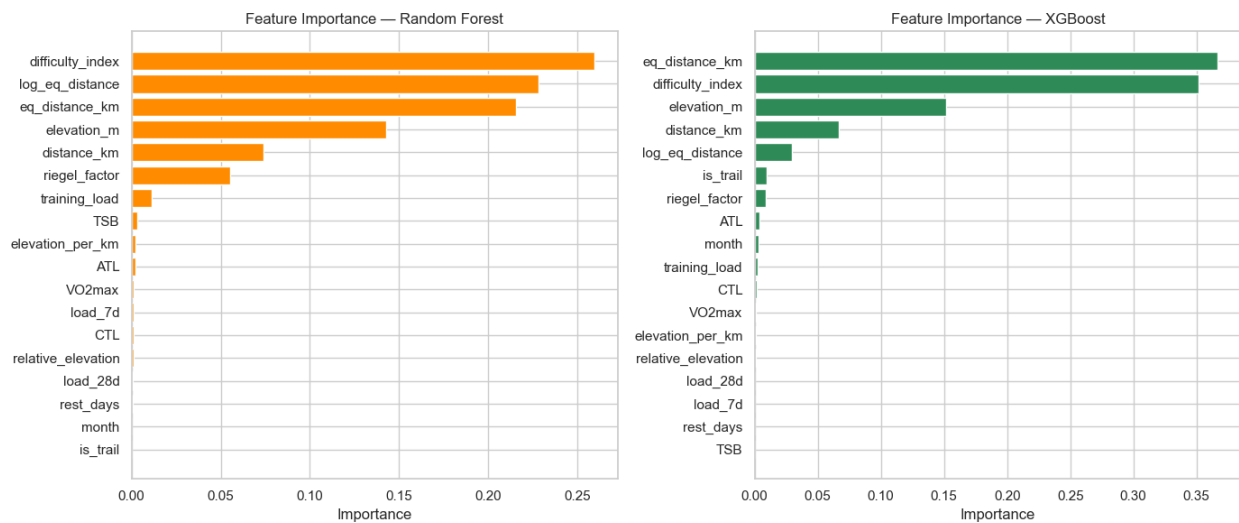


Figure 13: Feature importance

## 6.2 Models limitations

### Challenges posed by long distances and high elevation gain

As we have seen, the best model is the ML model. However, it is not reliable outside the training distribution. If we need to predict unknown runs, the predicted values will therefore be inaccurate.

To address this, two limitations have been imposed on the use of the ML model for pace prediction:



- DIST\_MAX\_ML = the maximum distance observed in the training dataset
- DENI\_MAX\_ML = maximum elevation difference observed in the training dataset

Outside these limits, Riegel's model is used for the prediction, as it incorporates fatigue variables that better account for long distances.

### **Calibration bias on mid-distances**

Overall, the model is well calibrated, though it exhibits a slight bias. However, there is a significant amount of data in the 10–15 km range where the overestimation is more pronounced. Furthermore, the relatively small size of the dataset limits its statistical robustness.

### **Lack of long distances run**

Currently, the results validate the modeling approach, but also point to a clear avenue for improvement: by enriching the dataset with more long-distance runs and greater variability in fitness levels, physiological features could play a more significant role, and their potential could be better harnessed.

## **7. Conclusion and outlook**

This project demonstrates that a personal race pace predictor can be built entirely from an individual athlete's Garmin training history, without relying on any external dataset or generic performance tables. By combining careful feature engineering, in particular the Riegel-based fatigue features and the TRIMP-derived training load indicators, with a gradient boosting model, the system achieves a mean absolute error of 3 minutes and 20 seconds and an  $R^2$  of 0.973 on the test set, making it a practically useful tool for race planning.

Several limitations must be acknowledged. The model performs best on road runs between 5 and 40 km, which represent the bulk of the training data. Predictions for distances beyond 68.5 km or elevation gains above 1,864 m rely on Riegel's extrapolation formula rather than the ML model, introducing greater uncertainty. A mild calibration bias of –1.9 minutes was also observed, particularly on runs in the 10–15 km range, which are underrepresented in the dataset. Finally, physiological state features such as ATL, CTL, TSB, and VO2max contribute only marginally to the model's predictions, suggesting that their full potential has not yet been captured given the current dataset size.

Several avenues for improvement are worth pursuing. Enriching the dataset with more long-distance trail runs would extend the ML model's reliable range and allow physiological features to play a more significant role. Incorporating additional signals such as temperature, altitude, or sleep quality could further refine predictions. From a technical standpoint, a hyperparameter search and cross-validation strategy would provide more robust model selection. Finally, the Streamlit application could be extended with a training load dashboard and automatic Garmin synchronization, turning the predictor into a fully autonomous planning tool that updates itself after every workout.

## 8. Annexes

### 8.1 List of features

| Feature            | Definition                                                                                                                                        | Scale          |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| distance_km        | Total distance of the run                                                                                                                         | Km             |
| eq_distance_km     | Distance adjusted for elevation: distance_km + elevation_m / 100. Converts trail difficulty into a road-equivalent distance                       | Km             |
| elevation_m        | Total positive elevation gain (D+) of the run                                                                                                     | M              |
| elevation_per_km   | Elevation gain per kilometer: elevation_m / distance_km. Measures terrain steepness independently of total distance                               | M              |
| is_trail           | Binary flag indicating race type: 0 = road, 1 = trail                                                                                             | 0 or 1         |
| Month              | Calendar month of the run (1 = January, 12 = December). Captures potential seasonal effects on performance                                        | 1-12           |
| riegel_factor      | distance_km <sup>1.06</sup> . Based on Riegel's formula, captures the progressive pace slowdown as distance increases                             | Dimensionless  |
| difficulty_index   | eq_distance_km <sup>1.06</sup> . Combines equivalent distance and the Riegel exponent — rises steeply for long mountain races                     | Dimensionless  |
| log_eq_distance    | Natural logarithm of eq_distance_km. Linearises the non-linear distance–duration relationship for linear models                                   | Dimensionless  |
| relative_elevation | elevation_m / distance_km. Measures terrain intensity independently of total distance                                                             | M/km           |
| ATL                | Acute Training Load — 7-day rolling average of daily TRIMP. Represents short-term fatigue                                                         | TRIMP units    |
| CTL                | Chronic Training Load — 42-day rolling average of daily TRIMP. Represents long-term fitness baseline                                              | TRIMP units    |
| TSB                | Training Stress Balance — CTL - ATL. Positive = fresh, negative = fatigued                                                                        | TRIMP units    |
| load_7d            | Cumulative TRIMP over the past 7 days. Complements ATL with the total recent workload rather than the average                                     | TRIMP units    |
| load_28d           | Cumulative TRIMP over the past 28 days. Captures medium-term training volume                                                                      | TRIMP units    |
| rest_days          | Number of days since the last recorded activity. Proxy for acute recovery                                                                         | Days           |
| VO2max             | Maximum aerobic capacity as estimated by Garmin. Updated after each run and reflects the athlete's current aerobic fitness level                  | mL/kg/min      |
| training_load      | Garmin's native training load score from the last run before the target date. Captures Garmin's own intensity weighting across all activity types | Arbitrary unit |

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## Statement

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